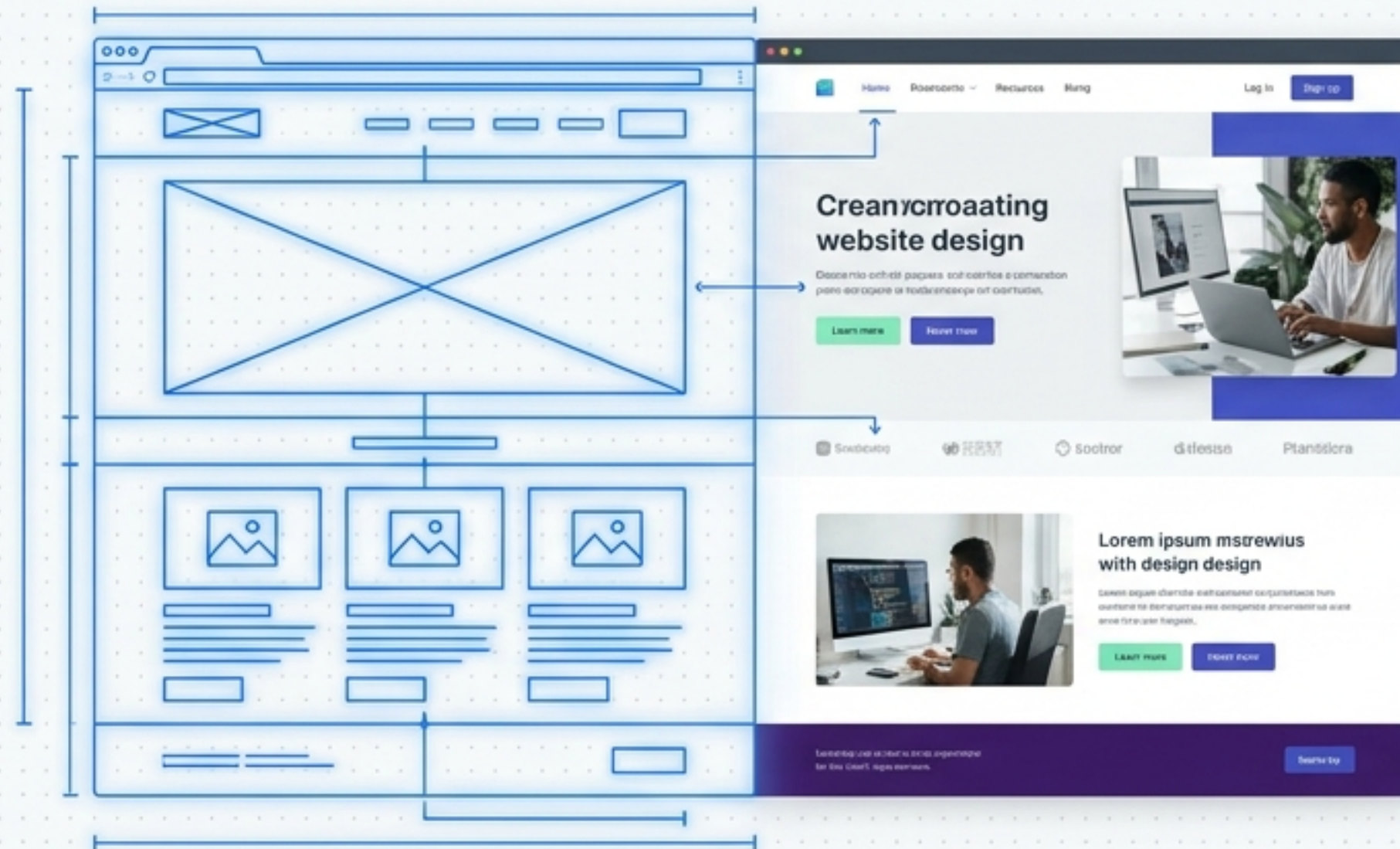


Introduction to Fullstack Web Development & HTML

Building the Blueprint for the Web.



Blueprint Legend

TEACHER NOTES

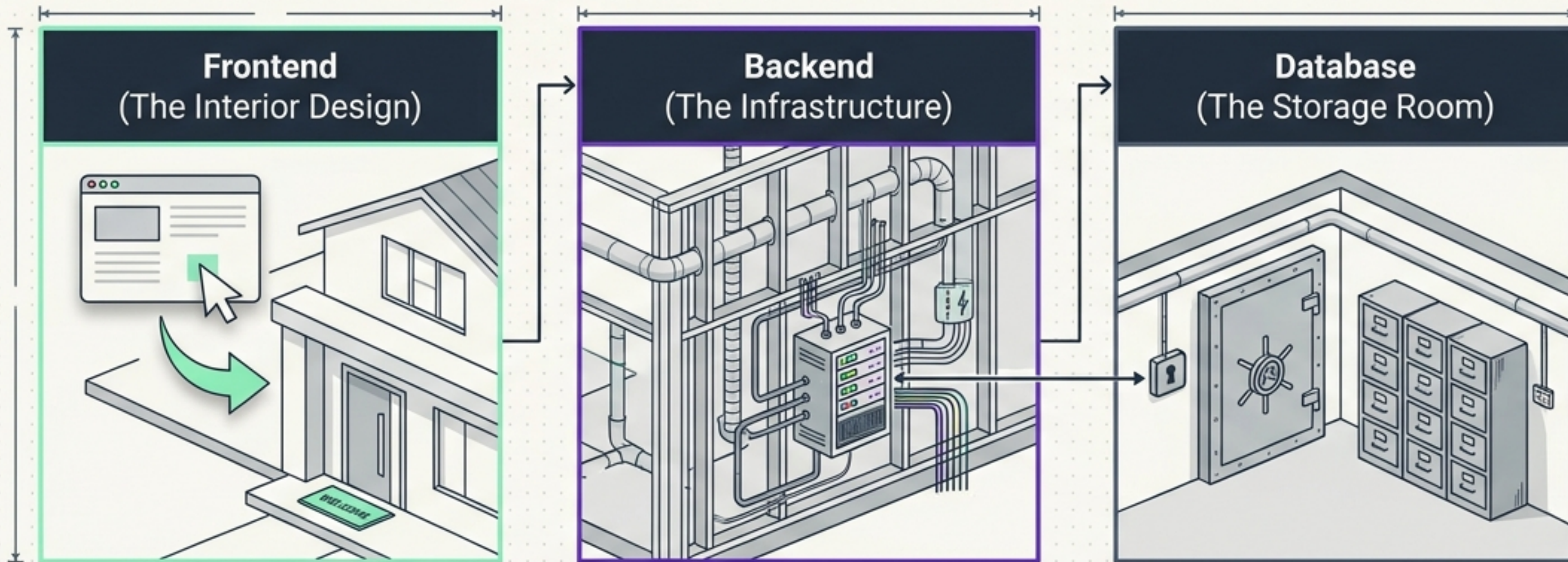
Welcome to Day 1. Today, we demystify the web. You will transition from being a consumer of websites to a creator. By the end of these 3 hours, you will have built the foundational blueprint of a real application.

STUDENT Q&A

Do I need any previous coding experience for this class? (Answer: None at all. We start from absolute zero today.)

The Fullstack Ecosystem: Building the House

Fullstack web development bridges the gap between the user's screen and the server room.



- Client-side (What the user sees)
- **HTML**: Structure / Framing
- **CSS**: Style / Paint
- **JavaScript**: Interactivity / Electricity

- Server-side logic
- Handles requests & routing
- Example: *Node.js, Python*

- Permanent data storage
- User accounts & inventory
- Example: *PostgreSQL, MongoDB*

IMPORTANT TERM: A Fullstack Developer works across all three of these pillars.

Blueprint Legend

TEACHER NOTES:

If a web app is a house, **HTML** is the wooden frame. You can't paint (**CSS**) without a frame. Today, we are carpenters building that frame.

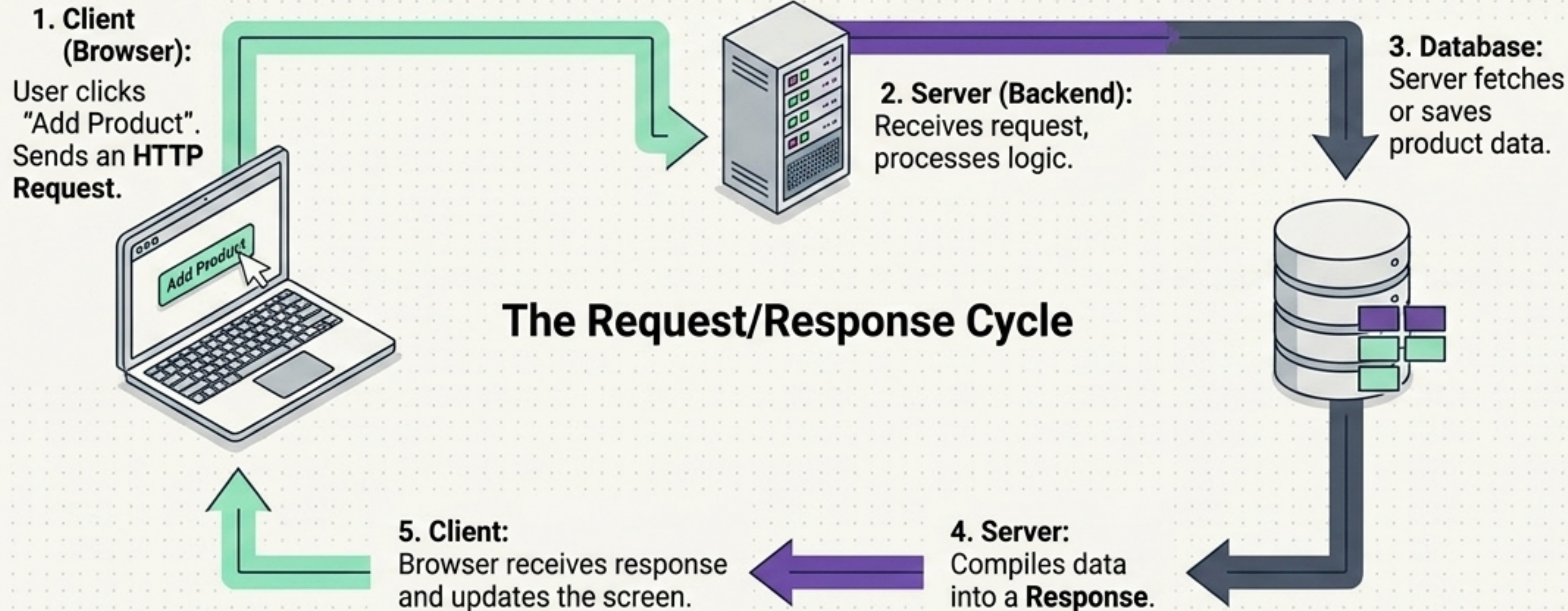
STUDENT Q&A:

If I only want to design websites, do I need to learn Backend?

How Web Applications Communicate

Websites are continuous conversations between a user's computer and remote servers.

The Request/Response Cycle



WHAT IS DEPLOYMENT? Moving your code from your local computer (where only you can see it) to a live server on the internet (where the world can see it).

Blueprint Legend

TEACHER NOTES:

Every time you click a button on Amazon, this exact loop happens in milliseconds. Deployment is how we put our 'house' on a public street.

STUDENT Q&A:

Where exactly does the database live?

Our Destination: The Product Inventory Management System

We learn best by building. This course builds a complete web application.

Blueprint Legend

TEACHER NOTES:

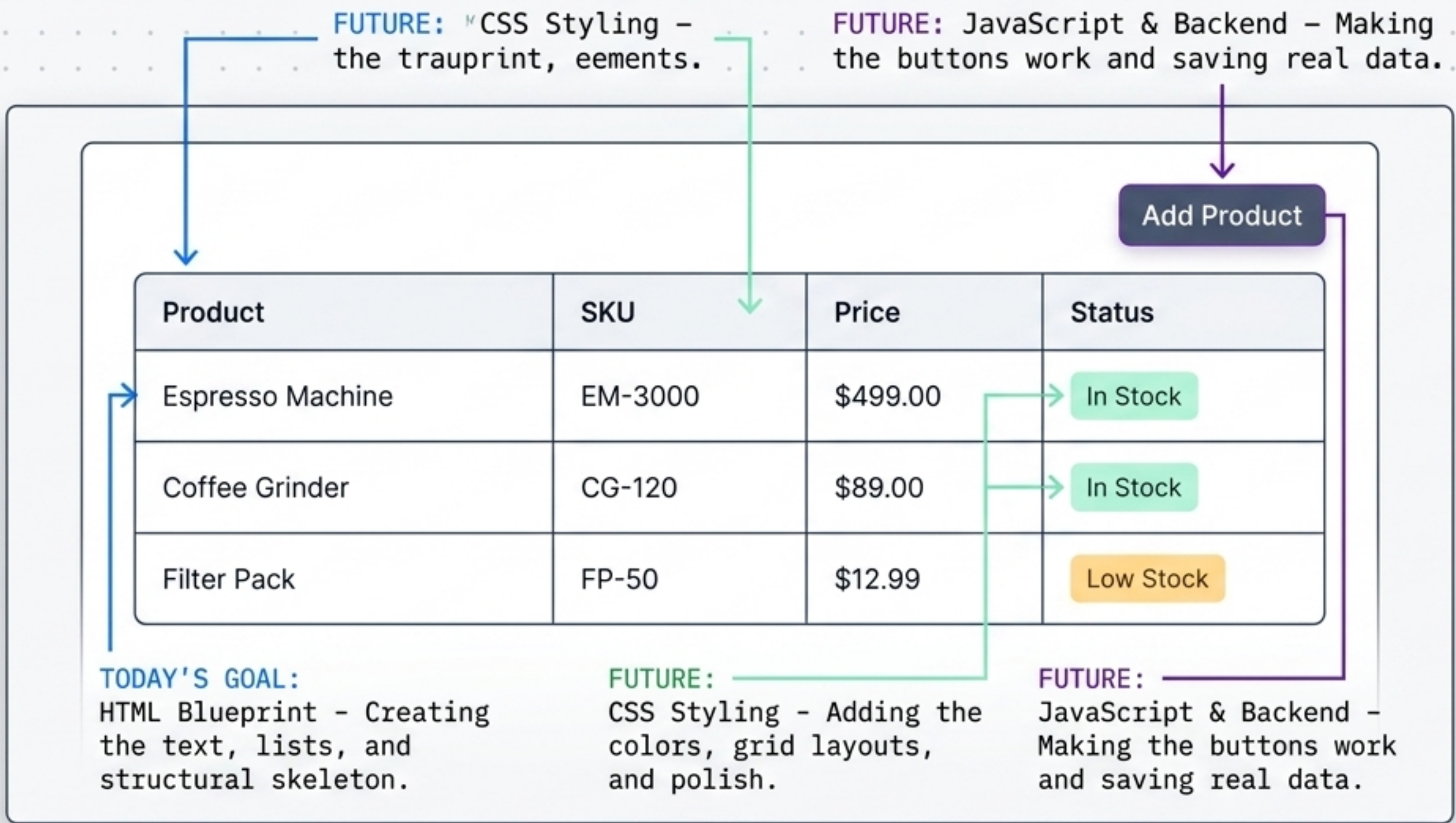
This is our final destination.

It looks complex, but we will build it piece by piece.

Today, we lay the foundation.

STUDENT Q&A:

Will this system actually be able to save real products by the end of the course?



What is HTML? The Language of Structure

HyperText Markup Language uses “markup” to annotate text so the browser knows how to display it.

Blueprint Legend

TEACHER NOTES:

HTML is not a programming language; it does not do math or logic.

It simply labels content:
“This is a heading”,
“This is a paragraph”.

STUDENT Q&A:

Does it matter if I
Does it matter if I use
uppercase letters for
tags, like <P>?

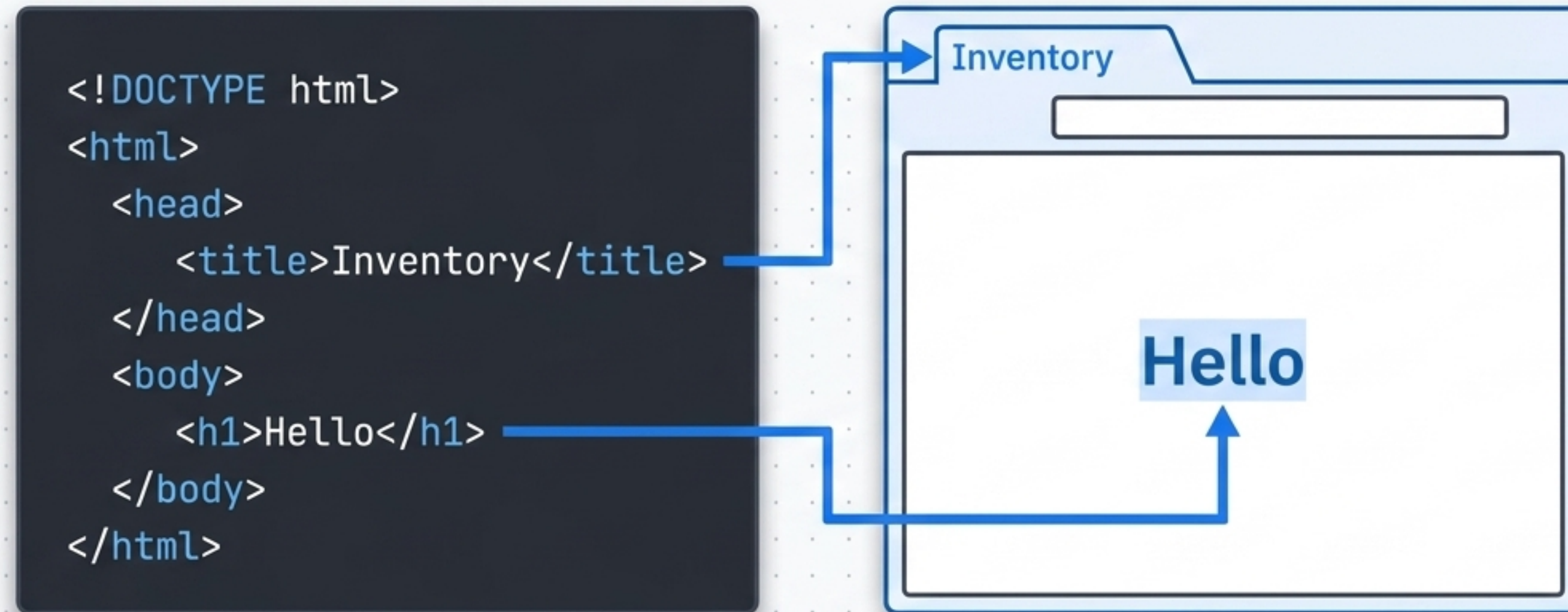


IMPORTANT TERMS:
Tag, Element,
Markup.

Opening Tag + Content + Closing Tag = **HTML Element**

The Basic HTML Document: Head vs. Body

Every valid HTML5 document requires a specific skeleton that splits invisible metadata from visible content.



Blueprint Legend

TEACHER NOTES:

Think of the Head as the brain—thinking, settings, identity. The Body is the physical form—what you actually see and touch.

STUDENT Q&A:

If metadata in the head is invisible, why do we need it?

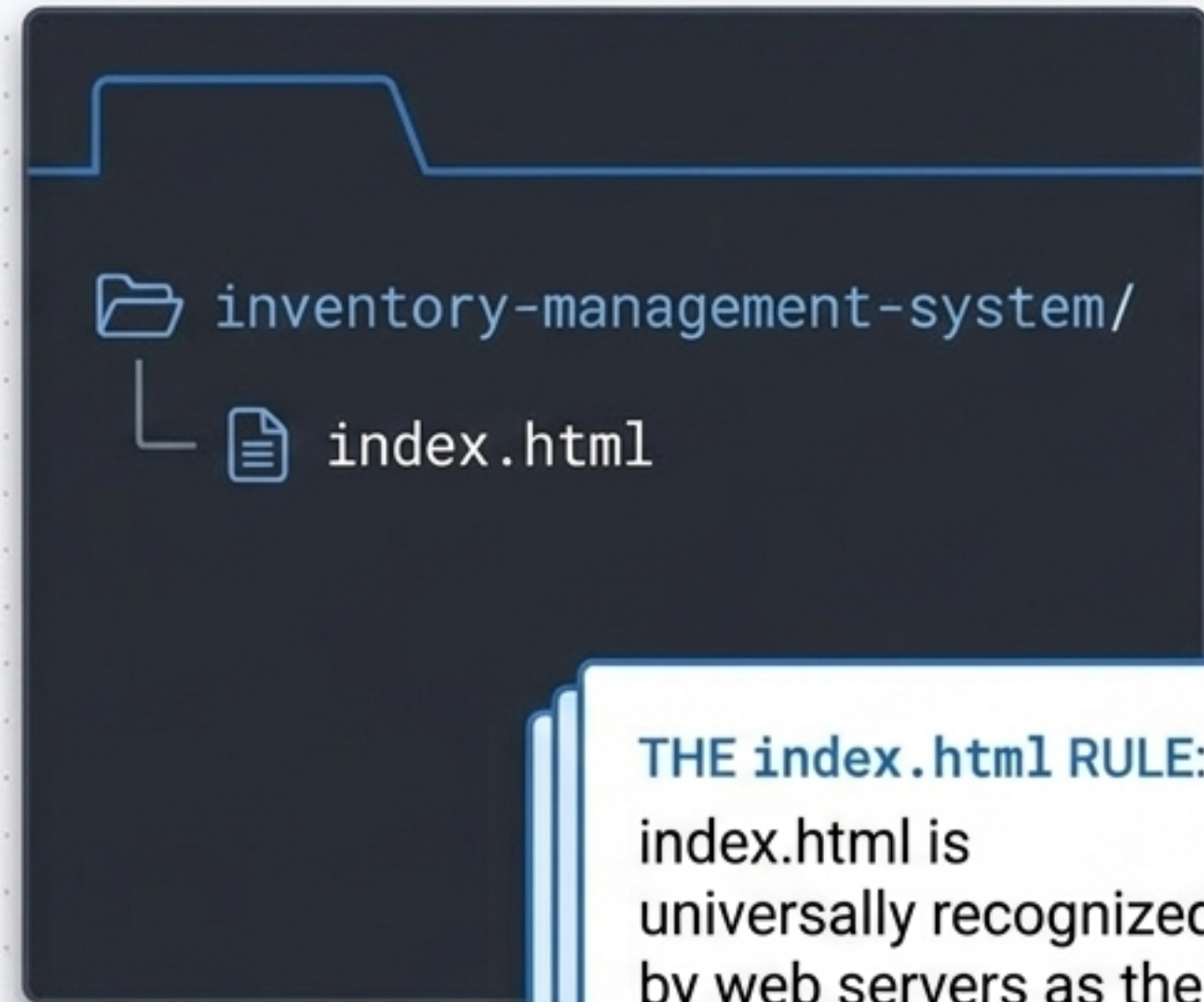
`<!DOCTYPE html>` : Tells browser to use modern HTML5 standards

`<head>` : Invisible metadata & settings

`<body>` : Visible physical content

Environment Setup: Dealing with Files

A website is just a folder of files on your computer. Web servers are strict about how you name them.



THE `index.html` RULE:

`index.html` is universally recognized by web servers as the default homepage.

Naming Conventions



DO

- my-project
- index.html
- about-us.html

(Lowercase, hyphens)



DON'T

- My Project
- Index.HTML
- about us.html

(Uppercase and spaces will break on live servers)

Blueprint Legend

TEACHER NOTES:

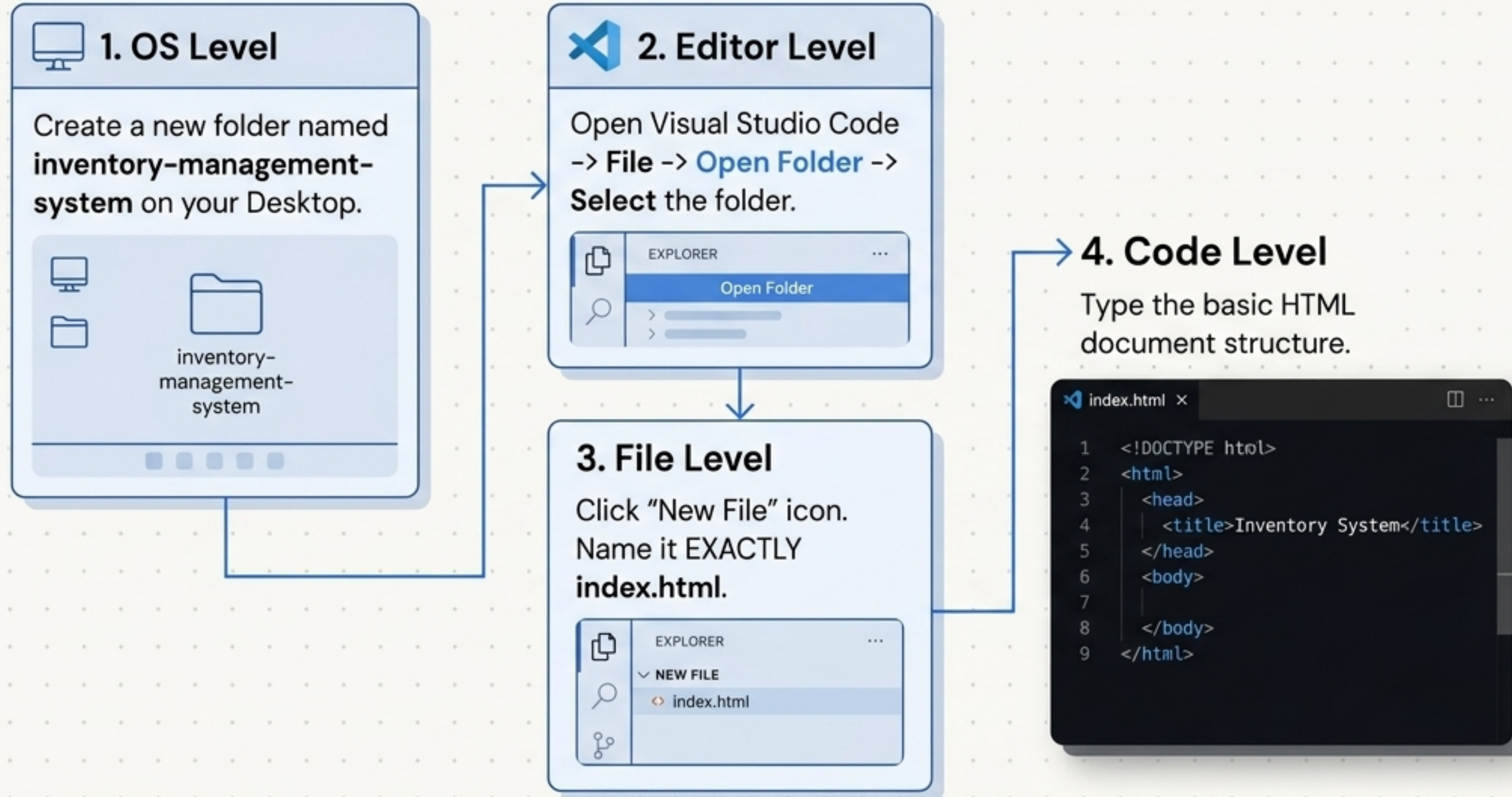
If you name a file with a space, the web server replaces the space with '%20', which breaks links. Always use lowercase and hyphens.

STUDENT Q&A:

Can I name my main file `home.html` instead of `index.html`?

Workshop: Project Setup (Steps 1–4)

Create your project folder and write the mandatory HTML skeleton.



Blueprint Legend

TEACHER NOTES:

I will pause here and walk around the room. Ensure your file is named exactly **index.html** with no capital "I".

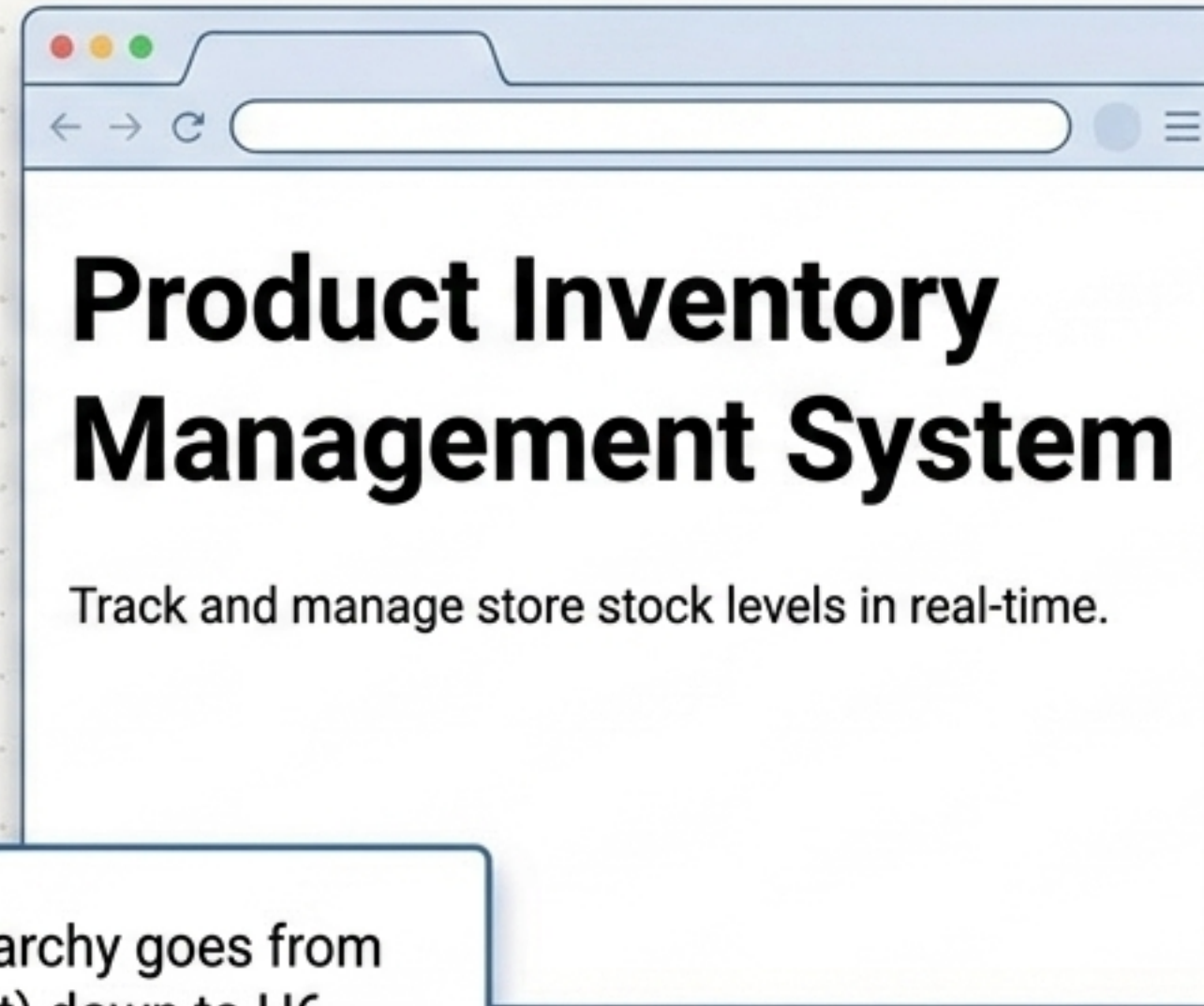
STUDENT Q&A:

My VS Code asks "Do you trust the authors of the files in this folder?" What do I click?
(Answer: Yes, because you are the author!)

Workshop: Adding the Project Title (Step 5)

Inside the `<body>` tag, we add visible content using Headings (`<h1>`) and Paragraphs (`<p>`).

```
index.html x
<body>
  <h1>Product Inventory Management
  System</h1>
  <p>Track and manage store stock
  levels in real-time.</p>
</body>
```



IMPORTANT: Heading hierarchy goes from H1 (largest/most important) down to H6. You should only have one H1 per page.

Blueprint Legend

TEACHER NOTES:

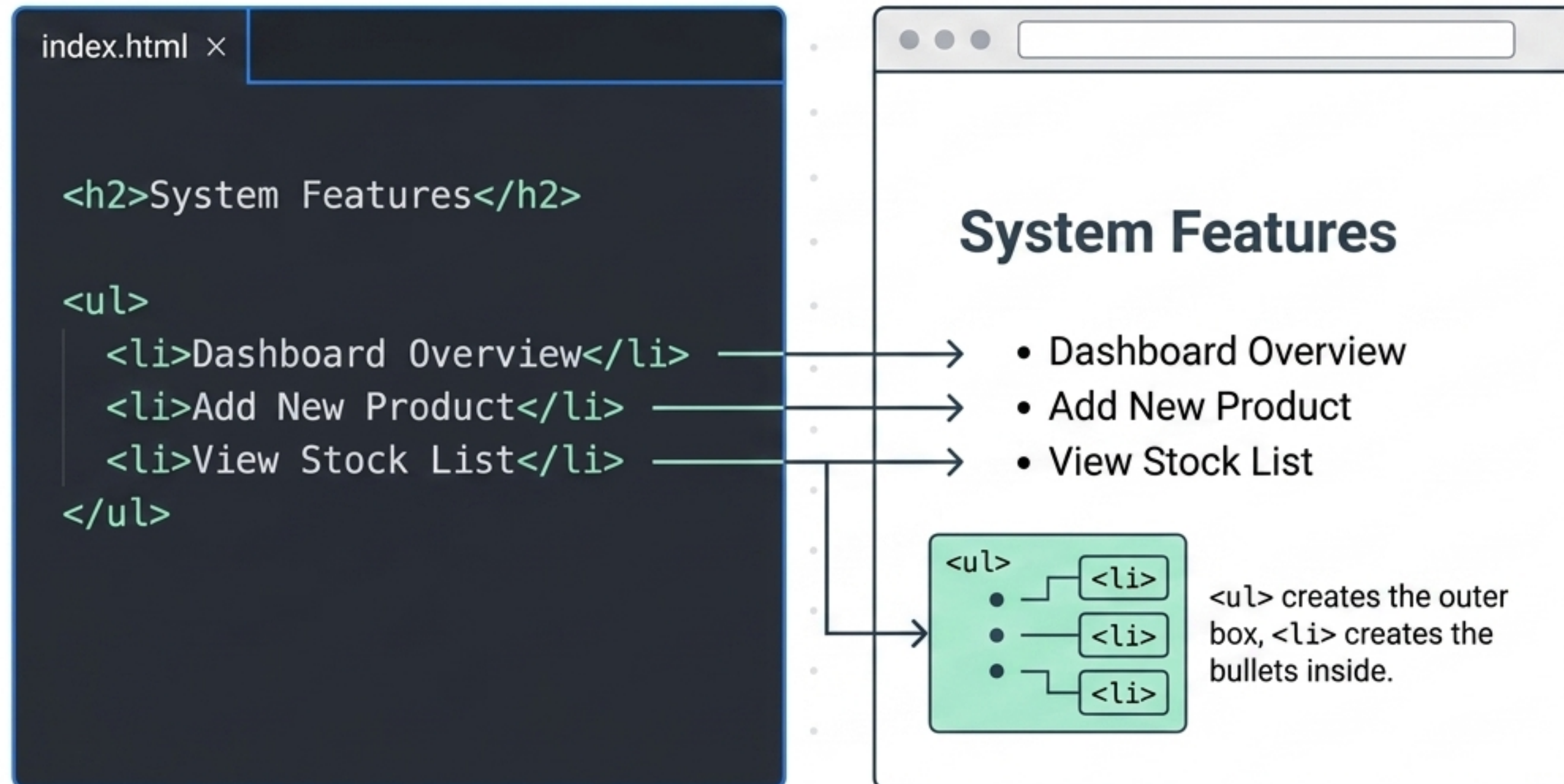
You should only ever have one `<h1>` per page. It acts like the title of a book. Paragraphs are for your standard reading text.

STUDENT Q&A:

Can I just use `<h1>` to make text bold, even if it's not a title?

Workshop: Adding Main Features (Step 6)

To group items cleanly, HTML uses Unordered Lists () which contain individual List Items ().



Blueprint Legend

TEACHER NOTES: Notice how we nest tags here. The tags live *inside* the tags. Indenting your code makes this relationship easy to read.

STUDENT Q&A: What if I wanted numbered steps instead of bullet points?

Workshop: Sample Products & Viewing (Steps 7–10)

Add sample data using sub-headings (<h3>) and execute the critical developer view loop.

CODE TO ADD

```
<h3>Espresso Machine</h3>
<p>SKU: EM-8991-A | Quantity: 12</p>
```

1. SAVE



Press **Ctrl+S** or **Cmd+S**.
A white dot on the VS Code tab means unsaved!

2. LOCATE



Open your OS File Explorer / Finder.

3. VIEW



Double-click **index.html** to open it in Chrome.

EXPECTED RESULT (Unstyled Browser View)



BLUEPRINT LEGEND

TEACHER NOTES:

This is the core developer loop: Write code, Save, Refresh the browser, Check the result. If you don't save, the browser won't see changes.

STUDENT Q&A:

Why doesn't it look like the pretty wireframe from the start of class?

(Answer: We only wrote **HTML structure**. We need **CSS** for the paint and polish!)

Common Beginner Mistakes

Even experienced developers make typos. Knowing how to spot these two errors will save you hours.

Mistake 1: Forgetting to Close Tags

✗ Wrong

```
<p>This is a paragraph
```

The browser gets confused and might make everything after it part of the paragraph.

✓ Right

```
<p>This is a paragraph</p>
```

Mistake 2: Incorrect Nesting (The "Russian Doll" Rule)

✗ Wrong

```
<p><h2>Welcome</h2></p>
```

Block elements like headings cannot go inside paragraphs.

✓ Right

```
<h2>Welcome</h2>  
<p>Intro text.</p>
```



Blueprint Legend

TEACHER NOTES:

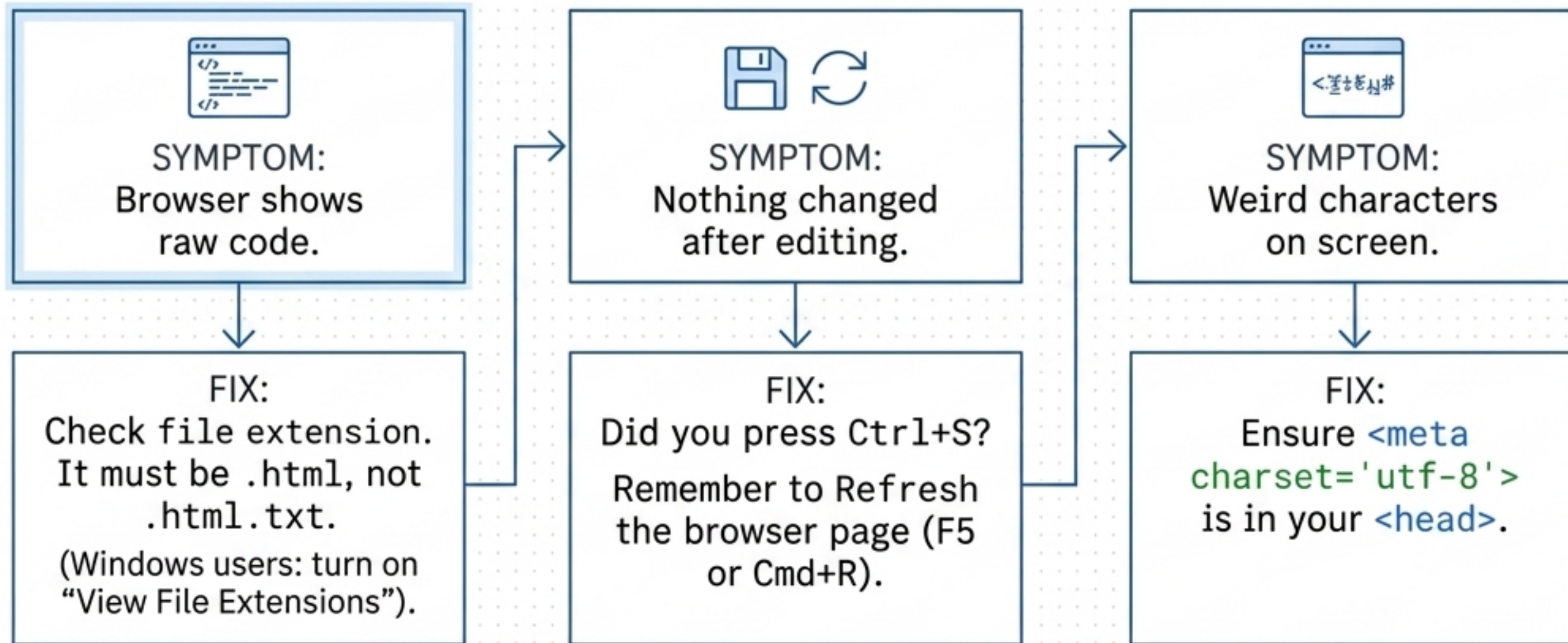
Browsers try to guess what you meant if you make a mistake, but eventually, they guess wrong and your layout breaks. Always close your tags.

STUDENT Q&A:

Will my code editor tell me if I forget to close a tag?

Troubleshooting Your Code

If your webpage looks wrong, follow this debugging checklist.



IMPORTANT TERM: **Debugging**

The process of identifying and removing errors from code.

Blueprint Legend

TEACHER NOTES:

Professional developers spend more time debugging than writing new code. Getting comfortable with a broken page is a superpower.

STUDENT Q&A:

Is there a tool that checks my code for me automatically?

Student Practice Task: Expand the Inventory

Apply the tags we just learned to expand our blueprint independently.

Task Briefing UI

OBJECTIVE

Add three new products to the inventory list independently.

REQUIREMENTS CHECKLIST

- ☐ Use an `<h2>` tag for a new section called 'New Arrivals'.
- ☐ Use a `<ul>` and `<li>` to create a bulleted list of 3 new products (e.g., Coffee Grinder, Milk Frother, Mugs).
- ☐ Add a short `<p>` description under the list.
- ☐ Save the file and refresh your browser to verify.



Blueprint Legend

TEACHER NOTES:

You have 10 minutes. Use the code we've already written as your reference guide. I'll walk around to help if you get stuck.

STUDENT Q&A:

Can I put the paragraph inside the list item?

Lesson Summary & Next Steps

You have successfully built the structural blueprint of a web application.

Recap

1. Fullstack Ecosystem spans Frontend, Backend, and Database.
2. HTML acts as the structural frame, not the paint.
3. Project files must be lowercase, no spaces, starting with `index.html`.

Review Questions

1. Where does invisible metadata go? (`<head>` or `<body>`)?
2. Why do we use `<ul>` instead of `<p>` for navigation menus?

Next Steps

HOMEWORK: Read ahead on CSS Basics.

NEXT CLASS: We add CSS styling to our inventory system to bring the blueprint to life.

Blueprint Legend

TEACHER NOTES: Great job today. You took a blank text file and turned it into a web document. Next time, then, we bring in CSS and give this blueprint some life.

STUDENT Q&EA: Where can I find the CSS reading for homework?

STUDENT Q&A: Where can I find the CSS reading material for homework?